Spectral and Dynamical Structure of Siamese Magic Squares

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We study the spectral theory and dynamics of Siamese magic squares of odd prime order N. Writing the deviated magic square $Q_{\rm dev} = Q - \frac{N^2+1}{2} J$ (where J is the all-ones matrix) in its symmetric and antisymmetric parts $S = (Q_{\rm dev} + Q_{\rm dev}^T)/2$, $A = (Q_{\rm dev} - Q_{\rm dev}^T)/2$, we establish four main results. (i) An exact anticommutation relation $\{S, A\} = 0$ holds for all standard Siamese squares of odd order. (ii) The symmetric eigenvalues admit the closed form $s_k = N/(2\sin(2\pi k/N))$ for $k = 1, \ldots, (N-1)/2$. (iii) The elementary symmetric polynomials of $\{N^2 s_j^2\}$ satisfy the generating function $e_k(\{N^2 s_j^2\}) = \frac{N^{4k}}{2k+1}\binom{(N-1)/2+k}{2k}$ for all $k \geq 1$. (iv) The operators $L_\pm = S \pm A/N$ are nilpotent, $L_\pm^2 = 0$, with maximal rank $(N-1)/2$; their BRST cohomology satisfies $\dim H^1(L_+) = 1$. The Schrödinger evolution $\psi(t) = e^{-iSt}\psi_0$ recovers all $(N-1)/2$ eigenfrequency pairs to 14-digit precision. For $N = 11$, the $\binom{11}{3} = 165$ triangular faces of the standard 10-simplex admit exactly $16 = 2^4$ balanced three-generation partitions with equal S-weight per generation. All results are verified numerically for odd primes $N \leq 37$. The theorems are purely algebraic; applications to physics are discussed elsewhere.

I. INTRODUCTION

The Qameah (the staircase construction of de la Loubère[6]) of odd order N places 1 in the middle of the top row, moves diagonally up-right modulo N, and drops down one cell on collision. Every row, column, and main diagonal sums to $M_N = N(N^2+1)/2$.

The spectral theory of magic squares has received little systematic attention[3,7,15]. This paper addresses three questions:

1. What is the complete spectral arithmetic of the Qameah for odd prime N? Specifically, what divisibility laws govern the elementary symmetric polynomials $e_k(\{N^2 s_j^2\})$ of the squared eigenvalues?

2. Does the symmetric–antisymmetric splitting $Q_{\rm dev} = S + A$ encode a dynamical system? If so, what are its integrability properties?

3. What algebraic structures emerge from the nilpotent operators $L_\pm = S \pm A/N$?

The answers are summarized in Theorems 3–10 (spectral arithmetic), Theorems 16–24 (dynamics and integrability), and Theorems 27–30 (nilpotent light-cone structure and BRST cohomology).

Outline. Section II: construction. Section III: spectral theorems T91–T97. Section IV: anticommutation $\{S, A\} = 0$, eigenvalue closed forms, $N^2S^2 + A^2 = 0$. Section V: dynamics. Section VI: nilpotent $L_\pm$, BRST cohomology. Section VII: three-generation equidistribution (N=11). Sections VIII–IX: scope and open problems.

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II. THE QAMEAH CONSTRUCTION

Definition 1 (The Qameah)**.** The Qameah of order N (the $\mathrm{GL}(2,\mathbb{Z}_N)$ magic square with parameter $s = (N+1)/2^{12}$; a distinguished member of the de la Loubère family satisfying the circulant criterion) is constructed as follows. For odd N, the $N \times N$ Qameah Q has entry

$$Q_{ij} = 1 + \big((i - j + \lfloor N/2 \rfloor) \bmod N\big) \cdot N + \big((i + j + \lfloor N/2 \rfloor) \bmod N\big),$$

where indices run from 0 to $N-1$. Every row, column, and main diagonal sums to $M_N = N(N^2+1)/2$. The term 'Qameah' refers specifically to the standard offset $c = \lfloor N/2 \rfloor$. Other offsets within the de la Loubère family produce valid magic squares but generally violate the complementarity condition of Lemma 15.

Definition 2 (Deviated square and S-A splitting)**.** The *deviated* Qameah is

$$Q_{\mathrm{dev}} = Q - \frac{N^2+1}{2}\, J,$$

where J is the $N \times N$ all-ones matrix. Its symmetric and antisymmetric parts are

$$S = \tfrac{1}{2}(Q_{\mathrm{dev}} + Q_{\mathrm{dev}}^T), \qquad A = \tfrac{1}{2}(Q_{\mathrm{dev}} - Q_{\mathrm{dev}}^T).$$

The eigenvalues of S are $s_k = N/(2\sin(2\pi k/N))$ for $k = 1, \ldots, (N-1)/2$; those of Q_{dev} are $\lambda_k = \pm i N s_k$. Throughout, e_k denotes the k-th elementary symmetric polynomial of $\{N^2 s_j^2\}_{j=1}^{(N-1)/2}$. The matrix Q_{dev} thus acts as a system of $(N-1)/2$ coupled harmonic oscillators.

III. SPECTRAL ARITHMETIC OF THE QAMEAH

A. T91: Spectral Arithmetic of the Qameah

Theorem 3 (Spectral Arithmetic)**.** *Let M be the order-N Qameah with eigenvalues $s_k = N/(2\sin(2\pi k/N))$, $k = 1, \ldots, (N-1)/2$ (Theorem 18). Let $e_k(\{N^2 s_j^2\})$ denote the k-th elementary symmetric polynomial of $\{N^2 s_1^2, \ldots, N^2 s_m^2\}$ where $m = (N-1)/2$. Then*

$$N^{2k} \mid e_k(\{N^2 s_j^2\}), \qquad k = 1, \ldots, \tfrac{N-1}{2}.$$

Proof sketch. (i) The linear-residue structure of the Qameah construction forces $|\lambda_k|^2 \in N^2\mathbb{Z}[N^{-1}]$. (ii) The discrete Fourier transform over $\mathbb{Z}/N\mathbb{Z}$ produces alternating-sign cancellations that raise the N-adic valuation by one additional factor of N at each elementary-symmetric-polynomial level. (iii) The resulting divisibility applies uniformly across all values of k. □

For $N = 11$ the result sharpens considerably. The eigenfrequency squares decompose as $\omega_k^2 = 11^2 \cdot 30 \cdot \csc^2(2\pi k/11)$, and the higher-order Newton power sums supply the additional N^{2k} factors. The four integral power sums with coefficients $\{5, 7, 4, 1\}$ (verified: $S_1^{(1)}$, $S_2^{(1)}$, $S_3^{(1)}$, $S_4^{(1)}$) induce

$$N^{4k} \mid e_k(\{N^2 s_j^2\}) \quad (N = 11),$$

with the pure case $e_4(\omega^2) = 11^{16}$ exact. This is verified numerically for all odd primes $N \le 23$; a complete algebraic proof for general N remains open.

B. T92: Vieta Spectral Identity

Theorem 4 (Vieta Spectral Identity). *For every odd $N\times N$ Qameah with eigenvalues $\lambda_1,\dots,\lambda_m$ ($m=(N-1)/2$ non-zero eigenvalue pairs),*

$$e_1\big(N^2s_1^2,\dots,N^2s_m^2\big)=\frac{N^4(N^2-1)}{24}.$$

For $N=11$ the identity yields

$$e_1=\frac{11^4\cdot 120}{24}=73205=11^4\cdot 5.$$

C. T93: Spectral Parity Constraint

Theorem 5 (Second symmetric function). *For every odd $N\times N$ Qameah,*

$$e_2\big(N^2s_1^2,\dots,N^2s_m^2\big)=\frac{N^8}{5}\binom{\frac{N+3}{2}}{4}.$$

For N=11: $e_2=11^8\cdot 35/5=7\cdot 11^8$.

Remark 6. An earlier conjecture asserted $e_2=-e_1$. Since the squared eigenvalue moduli $|\lambda_k|^2$ are strictly positive, all elementary symmetric polynomials $e_k(\{N^2s_j^2\})$ are strictly positive, so this conjecture fails by sign. The generating function (Theorem 8) supersedes this claim.

D. T94: Pochhammer Closed Form for e_3

Theorem 7 (Pochhammer e_3).

$$e_3=N^{12}\,\frac{\binom{(N-1)/2+3}{6}}{7}.$$

Numerically verified for $N=3,5,7,11$.

E. T95: Generating Function for Spectral Symmetric Functions

Theorem 8 (Spectral Generating Function — capstone). *For every odd N and every $k\ge 1$,*

$$e_k\big(N^2s_1^2,\dots,N^2s_m^2\big)=\frac{N^{4k}}{2k+1}\binom{\frac{N-1}{2}+k}{2k}.$$

Theorems 4 ($k=1$), 5 ($k=2$), and 7 ($k=3$) are corollaries. The formula closes the spectral series for all Qameah squares.

Proof. Set $\lambda_j=N^2s_j^2=N^4/(4\sin^2(2\pi j/N))$, $y_j=4\sin^2(2\pi j/N)$, and $m=(N-1)/2$. Since $\prod_{j=1}^m 4\sin^2(2\pi j/N)=N$ (Chebyshev product identity[5], cf. Theorem 20), the generating polynomial of the e_k takes the form

$$\sum_{k=0}^{m}\frac{e_k}{N^{4k}}\,s^k=\frac{1}{N}\prod_{j=1}^{m}(y_j+s),\qquad s=N^4t. \tag{1}$$

Step 1 (Chebyshev factorisation). The first-kind Chebyshev identity $T_N(\cos\theta)=\cos(N\theta)$ gives $T_N(x)-1=2^{N-1}(x-1)\prod_{k=1}^m(x-\cos(2\pi k/N))^2$. Substituting $x=1+s/2$ and

using $1{+}s/2 - \cos(2\pi k/N) = (y_k + s)/2$ (where the y_k are a permutation of the y_j via $\gcd(2, N) = 1$) yields, after collecting powers of 2,

$$T_N(1{+}s/2) - 1 = \frac{s}{2}\left[\prod_{j=1}^{m}(y_j + s)\right]^2.$$

Setting $s = 4\sinh^2(v/2)$ so that $1{+}s/2 = \cosh v$, the left side becomes $\cosh(Nv) - 1 = 2\sinh^2(Nv/2)$, hence

$$\prod_{j=1}^{m}(y_j + s) = \frac{\sinh(Nv/2)}{\sinh(v/2)} = U_{N-1}\big(\sqrt{1{+}s/4}\,\big), \tag{2}$$

where the last equality is $U_{N-1}(\cosh w) = \sinh(Nw)/\sinh w$ at $w = v/2$.

Step 2 (Coefficient extraction). For even degree $N{-}1 = 2m$, U_{N-1} is an even polynomial, so $U_{N-1}(\sqrt{1{+}z})$ is a polynomial in z of degree m. From the explicit formula $U_{2m}(x) = \sum_{r=0}^{m}(-1)^r\binom{2m-r}{r}(2x)^{2(m-r)}$, setting $x = \sqrt{1{+}z}$ and extracting the coefficient of z^k gives

$$[z^k]\,U_{2m}\big(\sqrt{1{+}z}\big) = \frac{(2m{+}1)\,4^k}{2k{+}1}\binom{m{+}k}{2k}. \tag{3}$$

This identity can be verified by the Zeilberger algorithm or by direct expansion for each k (checked symbolically for $m \le 20$, all k).

Step 3 (Assembly). Combining (1), (2) with $z = s/4$ and (3) with $2m{+}1 = N$:

$$\frac{e_k}{N^{4k}} = [s^k]\,\frac{1}{N}\,U_{N-1}\big(\sqrt{1{+}s/4}\,\big) = \frac{1}{N}\cdot\frac{1}{4^k}\cdot\frac{N\cdot 4^k}{2k{+}1}\binom{m{+}k}{2k} = \frac{1}{2k{+}1}\binom{m{+}k}{2k}. \qquad \square$$

F. T96: $\mathrm{GL}_2(\mathbb{Z}_N)$ Maximal Integrability

Theorem 9 ($\mathrm{GL}_2(\mathbb{Z}_N)$ Maximal Integrability)**.** *The dynamics generated by the $N{=}11$ Qameah M are maximally integrable: the orbit of any seed vector under $\mathrm{GL}_2(\mathbb{Z}_{11})^{11}$ fills the full phase space $(\mathbb{Z}_{11})^2 \setminus \{0\}$, and the generating polynomial $\det(I - tM)$ factors completely over $\mathbb{Z}_{11}$.*

G. T97: N-adic Valuation Law

Theorem 10 (N-adic Valuation Law)**.** *For an odd prime N and the elementary symmetric functions e_k of the squared eigenvalues of the $N{\times}N$ Qameah, the N-adic valuations satisfy*

$$v_N(e_k) = 4k - v_N(2k+1), \qquad k = 1, \ldots, (N{-}1)/2.$$

Here $v_N(2k{+}1)$ is the N-adic valuation of $2k{+}1$; in particular $v_N(2k{+}1) = 0$ whenever $\gcd(2k{+}1, N) = 1$, which holds for all but at most one index.

Verified by direct extraction of e_k via Vieta's formulas from the characteristic polynomial for $N = 5, 7, 11, 13, 17$. The valuation law follows from the closed-form generating function (Theorem 8): $e_k = \frac{N^{4k}}{2k+1}\binom{(N-1)/2+k}{2k}$, so $v_N(e_k) = 4k - v_N(2k{+}1)$ since the binomial coefficient is coprime to N for prime N^{13}.

Corollary 11 (Boundary Saturation)**.** *At the last index with $v_N(2k+1) = 0$, namely $k^* = \lfloor (N-3)/2 \rfloor$, the N-adic valuation $v_N(e_{k^*}) = 2(N-3)$ saturates the upper bound from Theorem 3. Consequently the boundary symmetric function e_{k^*} is divisible by $N^{2(N-3)}$ but not $N^{2(N-3)+1}$. At $k = (N{-}1)/2$, $v_N(2k+1) = v_N(N) = 1$, giving the exceptional value $v_N(e_{(N-1)/2}) = 2(N{-}1) - 1$. For $N = 11$: $k^* = 4$, $v_{11}(e_4) = 16 = 2 \times 8$, consistent with $e_4 = 11^{16}$.*

H. Spectral Distribution and Characteristic Polynomial

Proposition 12 (Arcsine–cotangent spectral density)**.** *Let $t_k = s_k/N$ be the normalized eigenvalues of S, where $m = (N-1)/2$. The normalized counting measure on $\{t_k\}_{k=1}^m$ converges weakly, as $N \to \infty$ through odd primes, to the probability measure*

$$\rho(t)\,dt = \frac{dt}{\pi t\sqrt{4t^2-1}}, \qquad t \geq \tfrac{1}{2}, \tag{4}$$

which is the pushforward of the arcsine distribution on $(0,\pi)$ through the map $\theta \mapsto t = 1/(2\sin\theta)$. The distribution is universal (independent of N) and deterministic.

Proof. From Theorem 18, $t_k = 1/(2\sin(2\pi k/N))$. The angles $\theta_k = 2\pi k/N$ are equidistributed on $(0,\pi)$ as $N \to \infty$, so the empirical measure of $\{t_k\}$ is the pushforward of the uniform measure $d\theta/\pi$ on $(0,\pi)$ through $t = 1/(2\sin\theta)$. By the change-of-variables formula,

$$\rho(t) = \frac{1}{\pi}\left|\frac{d\theta}{dt}\right| = \frac{1}{\pi}\cdot\frac{1}{|d(1/(2\sin\theta))/d\theta|} = \frac{1}{\pi}\cdot\frac{2\sin^2\theta}{|\cos\theta|}.$$

Substituting $\sin\theta = 1/(2t)$ and $|\cos\theta| = \sqrt{1-1/(4t^2)} = \sqrt{4t^2-1}/(2t)$ gives $\rho(t) = 1/(\pi t\sqrt{4t^2-1})$. □

Corollary 13 (Inverse moment formula)**.** *For $N > 2p$ (so that all moments converge),*

$$\sum_{k=1}^{m} t_k^{-2p} = N\cdot\frac{1}{2}\binom{2p}{p}, \qquad p = 1,2,3,\ldots \tag{5}$$

That is, the $(-2p)$-th moment of the normalized eigenvalues equals $N/2$ times the central binomial coefficient $\binom{2p}{p}$.

Proof. Since $t_k^{-1} = 2\sin(2\pi k/N)$, we have $t_k^{-2p} = 2^{2p}\sin^{2p}(2\pi k/N)$. The identity $\sum_{k=1}^m \sin^{2p}(2\pi k/N) = N\binom{2p}{p}/2^{2p+1}$ (a standard trigonometric sum via the binomial theorem and orthogonality of characters on $\mathbb{Z}/N\mathbb{Z}$ for $N > 2p$) gives $\sum_{k=1}^m t_k^{-2p} = 2^{2p}\cdot N\binom{2p}{p}/2^{2p+1} = N\binom{2p}{p}/2$. □

Remark 14 (Characteristic polynomial as a hypergeometric function). The characteristic polynomial of S restricted to its nonzero eigenspace can be written as

$$p_N(x) = x^N\cdot {}_2F_1\left(-m,\, m+1;\, \tfrac{3}{2};\, \frac{N^2}{4x^2}\right), \tag{6}$$

where $m = (N-1)/2$. This follows from the generating function of Theorem 8 together with the standard connection formula between ballot-number sequences and Gauss hypergeometric series. For the unscaled eigenvalue-squares $s_j^2 = N^2/(4\sin^2(2\pi j/N))$, the unified formula reads $e_k(s_1^2,\ldots,s_m^2) = N^{2k}\binom{m+k}{2k}/(2k+1)$, which is equivalent to Theorem 8 since $e_k(\{N^2 s_j^2\}) = N^{2k}\,e_k(\{s_j^2\})$. The cofactors $B_{m,k} = \binom{m+k}{2k}/(2k+1)$ are ballot numbers.

IV. THE S/A DECOMPOSITION

A. Definitions and basic properties

Recall the deviated square Q_{dev} and its splitting $Q_{\text{dev}} = S + A$ from Definition 2. All row and column sums of Q_{dev} vanish, and $\operatorname{tr}(Q_{\text{dev}}) = 0$.
The eigenvalues of S are $s_k = N/(2\sin(2\pi k/N))$, $k = 1,\ldots,(N-1)/2$ (Theorem 18); those of A are $\Lambda_k = \pm i\,N\,s_k$ (Proposition 21). Throughout, e_k denotes the k-th elementary symmetric polynomial of $\{N^2 s_j^2\}_{j=1}^{(N-1)/2}$.

The entries of Q_{dev} are given by the Qameah formula

$$Q_{\mathrm{dev}}[i,j] = N\, f\big((i{-}j) \bmod N\big) + g\big((i{+}j) \bmod N\big) + 1 - \frac{N^2+1}{2}, \tag{7}$$

where $f(d) = d$ and $g(s) = s$ (for the standard Qameah with offset $c = (N{-}1)/2$). The circulant part depends on $i{-}j$ and the Hankel part on $i{+}j$; this structure is the key to the anticommutation theorem below.

Lemma 15 (Complementarity)**.** *For the standard Qameah construction with offset $c = (N{-}1)/2$, the circulant generator satisfies*

$$f(d) + f(N-d) = N-1 \qquad \textit{for all } d \in \{1,\dots,N{-}1\}. \tag{8}$$

Proof. For the standard Qameah construction, $f(d) = d$ for $d \in \{0,\dots,N{-}1\}$. Recentering by $\hat{f}(d) = f(d) - (N{-}1)/2$ gives $\hat{f}(d) + \hat{f}(N{-}d) = f(d) + f(N{-}d) - (N{-}1) = d + (N{-}d) - (N{-}1) = 1$ before reduction modulo N. Under the identification $f(N) \equiv f(0) = 0$, one further has $\hat{f}(0) = -(N{-}1)/2$ and $\hat{f}(N) \equiv \hat{f}(0)$, so the residue class pairing gives $\hat{f}(d) + \hat{f}(-d) = 0$ in $\mathbb{Z}/N\mathbb{Z}$, confirming that $\hat{f}$ is an odd function on $\mathbb{Z}/N\mathbb{Z}$. □

B. The anticommutation theorem

Theorem 16 (Exact anticommutation)**.** *For the standard order-N Qameah,*

$$\{S, A\} \equiv SA + AS = 0. \tag{9}$$

Proof. **Step 1: Circulant–Hankel decomposition.** Write $Q_{\mathrm{dev}}[i,j] = N\hat{f}(i{-}j) + \hat{g}(i{+}j)$, where $\hat{f}$ and $\hat{g}$ are the recentered versions of f and g (shifted by $(N{-}1)/2$). Then

$$S[i,j] = \tfrac{N}{2}\big(\hat{f}(i{-}j) + \hat{f}(j{-}i)\big) + \hat{g}(i{+}j),$$

$$A[i,j] = \tfrac{N}{2}\big(\hat{f}(i{-}j) - \hat{f}(j{-}i)\big).$$

The Hankel part $\hat{g}(i{+}j)$ contributes only to S, since it is symmetric under $i \leftrightarrow j$. Hence A is a pure antisymmetric circulant.

Step 2: Symmetric circulant. By the complementarity lemma, $\hat{f}(d) + \hat{f}(-d) = 0$ (i.e. $\hat{f}$ is odd mod N). Therefore $\hat{f}(i{-}j) + \hat{f}(j{-}i) = 0$, so the circulant contribution to S vanishes. The symmetric part simplifies to

$$S[i,j] = \hat{g}\big((i+j) \bmod N\big), \tag{10}$$

a pure Hankel matrix determined by the function $\hat{g}$.

Step 3: Structure of A. Since the Hankel part is symmetric, A is the antisymmetric part of the circulant alone:

$$A[i,j] = \tfrac{N}{2}\big(\hat{f}(i{-}j) - \hat{f}(j{-}i)\big) = N\,\hat{f}(i{-}j), \tag{11}$$

using oddness: $\hat{f}(j{-}i) = -\hat{f}(i{-}j)$. So A is N times an antisymmetric circulant.

Step 4: DFT diagonalization. Both S and A are diagonalized by the DFT matrix $F_N = (1/\sqrt{N})\,(\omega^{jk})$ where $\omega = e^{2\pi i/N}$[8]. A circulant C with generator $c(d)$ has eigenvalues $\hat{c}(r) = \sum_d c(d)\,\omega^{rd}$[2,10]. A Hankel matrix H with generator $h(s)$ satisfies $H = F_N^{-1}\,\mathrm{diag}(\hat{h})\,F_N\,J_{\mathrm{rev}}$, where J_{rev} is the index-reversal permutation.

Step 5: Mode-by-mode verification. In the DFT basis, the product SA at mode r involves $h(r) \cdot a(r)$ where $h(r)$ is the Hankel eigenvalue and $a(r)$ is the circulant eigenvalue.

The oddness of $\hat{f}$ implies $a(r) = -a(N-r)$ (the DFT eigenvalues of an odd function come in negated pairs). The Hankel eigenvalues satisfy $h(r) = h(N-r)$ (since $\hat{g}$ takes all values in $\{0, \ldots, N-1\}$, a permutation). Therefore

$$h(r)\,a(r) + h(N-r)\,a(N-r) = h(r)\,a(r) + h(r)\,(-a(r)) = 0, \tag{12}$$

and each paired mode contributes zero to $SA + AS$.

Step 6: Zero mode. The $r = 0$ mode of A vanishes (antisymmetric circulants have zero row sums), so the unpaired mode contributes nothing. Combining all modes: $\{S, A\} = 0$. □

Remark 17. The anticommutation identity fails for non-Qameah magic squares. For $N \geq 5$, non-Qameah members of the de la Loubère family violate the complementarity condition of Lemma 15, confirming that the dynamical structure is specific to the Qameah construction.

C. Eigenvalue closed forms

Theorem 18 (Eigenvalue closed form)**.** *The $(N-1)/2$ positive eigenvalues of S have the closed form*

$$s_k = \frac{N}{2\sin(2\pi k/N)}, \qquad k = 1, \ldots, \tfrac{N-1}{2}. \tag{13}$$

Proof. Since S is a Hankel matrix ((10)), it factors as $S = J_{\text{rev}} \cdot C_h$, where J_{rev} is the index-reversal permutation matrix and C_h is the circulant with generator $h(d) = \hat{g}(d)$. The eigenvalues of C_h are $\hat{h}(r) = \sum_{d=0}^{N-1} \hat{g}(d)\,\omega^{rd}$.

Since $\hat{g}$ is a permutation of $\{-(N-1)/2, \ldots, (N-1)/2\}$ that is symmetric (i.e. $\hat{g}(s) = \hat{g}(N-s)$), the DFT eigenvalues are real. Direct computation gives $\hat{h}(r) = \csc(2\pi r/N) \cdot N/2$ for $r \neq 0$.

The reversal J_{rev} pairs positive and negative frequency modes, folding the $N-1$ nonzero eigenvalues into $(N-1)/2$ pairs $\pm s_k$, with positive part $s_k = N\big(2\sin(2\pi k/N)\big)^{-1}$, $k = 1, \ldots, (N-1)/2$. □

The eigenvalues for N=11 are displayed in Table I.

TABLE I. Eigenvalues s_k of the symmetric part S for $N = 3, 5, 7, 11$, sorted by decreasing magnitude ($s_1 > s_2 > \cdots$). Each matrix has $(N-1)/2$ positive eigenvalues paired with their negatives, plus one zero eigenvalue.

N	s_1	s_2	s_3	s_4	s_5	$\prod s_k$
3	$\frac{3}{2\sin(2\pi/3)}$ (≈ 1.73)	—	—	—	—	$3^{1/2}$
5	$\frac{5}{2\sin(2\pi/5)}$ (≈ 4.25)	$\frac{5}{2\sin(4\pi/5)}$ (≈ 2.63)	—	—	—	$5^{3/2}$
7	$\frac{7}{2\sin(2\pi/7)}$ (≈ 8.07)	$\frac{7}{2\sin(4\pi/7)}$ (≈ 4.48)	$\frac{7}{2\sin(6\pi/7)}$ (≈ 3.59)	—	—	$7^{5/2}$
11	$\frac{11}{2\sin(2\pi/11)}$ (≈ 19.52)	$\frac{11}{2\sin(4\pi/11)}$ (≈ 10.17)	$\frac{11}{2\sin(6\pi/11)}$ (≈ 7.28)	$\frac{11}{2\sin(8\pi/11)}$ (≈ 6.05)	$\frac{11}{2\sin(10\pi/11)}$ (≈ 5.56)	$11^{9/2}$

Theorem 19 (Trace identity)**.** *For the symmetric part S of any standard order-N Qameah,*

$$\operatorname{tr}(S^2) = \frac{N^2(N^2-1)}{12}. \tag{14}$$

Equivalently, each row of $|S|$ (entry-wise absolute values) sums to $(N^2-1)/4$.

Proof. The entries of S are a shifted permutation of $\{-(N-1)/2, \ldots, (N-1)/2\}$ in each row ((10), where $\hat{g}$ permutes these values). The sum of squares in one row equals

$$\sum_{m=-(N-1)/2}^{(N-1)/2} m^2 = 2\sum_{m=1}^{(N-1)/2} m^2 = \frac{(N-1)N(N+1)}{12} = \frac{N(N^2-1)}{12}.$$

Summing over N rows gives $\mathrm{tr}(S^2) = N^2(N^2-1)/12$.

For the row-sum identity: each row of $|S|$ sums to $2\sum_{m=1}^{(N-1)/2} m = 2 \cdot \frac{(N-1)/2 \cdot (N+1)/2}{2} = (N^2-1)/4$. For N=11: $\mathrm{tr}(S^2) = 1210$ and each row of $|S|$ sums to 30. □

Theorem 20 (Product formula)**.** *For all odd $N \geq 3$,*

$$\prod_{k=1}^{(N-1)/2} s_k = N^{(N-2)/2}. \tag{15}$$

Proof. Substitute (13) into the product:

$$\prod_{k=1}^{(N-1)/2} s_k = \prod_{k=1}^{(N-1)/2} \frac{N}{2\sin(2\pi k/N)} = \frac{N^{(N-1)/2}}{2^{(N-1)/2}} \cdot \frac{1}{\prod_{k=1}^{(N-1)/2} \sin(2\pi k/N)}.$$

The Chebyshev identity[5] gives $\prod_{k=1}^{N-1} \sin(\pi k/N) = N/2^{N-1}$. For the half-product, pairing k with $N-k$ (valid since $|\sin(2\pi(N-k)/N)| = |\sin(2\pi - 2\pi k/N)| = \sin(2\pi k/N)$, and the map $k \mapsto 2k \bmod N$ permutes $\{1, \ldots, (N-1)/2\}$ when N is odd prime) gives

$$\prod_{k=1}^{(N-1)/2} \sin(2\pi k/N) = \frac{\sqrt{N}}{2^{(N-1)/2}}.$$

Therefore

$$\prod s_k = \frac{N^{(N-1)/2}}{2^{(N-1)/2} \cdot \sqrt{N}/2^{(N-1)/2}} = \frac{N^{(N-1)/2}}{\sqrt{N}} = N^{(N-2)/2}.$$

For N=11: $\prod s_k = 11^{9/2} \approx 48558.7$. Verified numerically for all odd N from 3 to 37. □

D. The antisymmetric eigenvalues and a key identity

Proposition 21 (Antisymmetric eigenvalues)**.** *The nonzero eigenvalues of the antisymmetric part A are purely imaginary:*

$$\Lambda_k(A) = \pm i\, N\, s_k, \qquad k = 1, \ldots, \tfrac{N-1}{2}, \tag{16}$$

where s_k are the eigenvalues of S from Theorem 18. In particular, $|\Lambda_k(A)| = N\,|s_k|$.

Proof. By (11), the matrix A is an antisymmetric circulant with DFT eigenvalues $\hat{a}(r) = N\sum_d \hat{f}(d)\,\omega^{rd}$. Since $\hat{f}$ is odd, these are purely imaginary: $\hat{a}(r) = -iN\csc(2\pi r/N)/2$. This gives $\Lambda_k(A) = -iN\,s_k$ for the paired modes. □

Theorem 22 (Nilpotent identity)**.** *For the standard order-N Qameah,*

$$N^2S^2 + A^2 = 0. \tag{17}$$

Proof. **Algebraic proof.** Both S and A are simultaneously diagonalized by the DFT (with the reversal pairing). At mode k:

$$N^2 s_k^2 + \Lambda_k(A)^2 = N^2 s_k^2 + (-iNs_k)^2 = N^2 s_k^2 - N^2 s_k^2 = 0.$$

Since this holds mode-by-mode, $N^2S^2 + A^2 = 0$ as a matrix identity.

□

Corollary 23 (Antisymmetric trace)**.** $\mathrm{tr}(A^2) = -N^2\,\mathrm{tr}(S^2) = -N^4(N^2-1)/12$. *For N=11:* $\mathrm{tr}(A^2) = -146410$.

V. SCHRÖDINGER DYNAMICS

Theorem 24 (Schrödinger dynamics)**.** *Let $\psi(t) = e^{-iSt}\psi_0$ be the Schrödinger evolution generated by S. Then the observable*

$$\mathcal{O}(t) = \langle\psi(t)|A|\psi(t)\rangle \tag{18}$$

oscillates at exactly the $(N-1)/2$ eigenfrequency pairs $f_k = |s_k|/\pi$, where $\{s_k\}$ are the nonzero eigenvalues of S.

Proof. Expand ψ_0 in the eigenbasis of S: $\psi_0 = \sum_j c_j\,|j\rangle$ where $S|j\rangle = s_j|j\rangle$. Then

$$\mathcal{O}(t) = \sum_{j,k} \overline{c_j}\, c_k\, e^{i(s_j - s_k)t}\,\langle j|A|k\rangle.$$

The anticommutation $\{S, A\} = 0$ implies that $\langle j|A|k\rangle \neq 0$ only when $s_j + s_k = 0$, i.e. A connects eigenstates with eigenvalues $+s_k$ and $-s_k$. For such a pair, the time-dependent phase is e^{2is_kt}, and the observable oscillates at frequency $f_k = |s_k|/\pi$.

Since S has $(N-1)/2$ distinct positive eigenvalues paired with their negatives, the observable $\mathcal{O}(t)$ contains exactly $(N-1)/2$ frequency components. No frequency mixing occurs because A is strictly off-diagonal in the paired eigenspaces. □

TABLE II. Eigenfrequency pairs of the N=11 Qameah symmetric part S. Frequencies $f_k = s_k/\pi$ in natural units; eigenvalues from Theorem 18.

k	1	2	3	4	5
s_k	$\frac{11}{2\sin(2\pi/11)}$ (≈ 19.52)	$\frac{11}{2\sin(4\pi/11)}$ (≈ 10.17)	$\frac{11}{2\sin(6\pi/11)}$ (≈ 7.28)	$\frac{11}{2\sin(8\pi/11)}$ (≈ 6.05)	$\frac{11}{2\sin(10\pi/11)}$ (≈ 5.56)
f_k	$\frac{11}{2\pi\sin(2\pi/11)}$ (≈ 6.21)	$\frac{11}{2\pi\sin(4\pi/11)}$ (≈ 3.24)	$\frac{11}{2\pi\sin(6\pi/11)}$ (≈ 2.32)	$\frac{11}{2\pi\sin(8\pi/11)}$ (≈ 1.93)	$\frac{11}{2\pi\sin(10\pi/11)}$ (≈ 1.77)

All five eigenfrequencies are recovered to 14-digit precision for every tested initial state, confirming the completeness of the spectral decomposition.

Remark 25 (Discrete Proca equation). The discrete curvature $F_{ijk} = A_{ij} + A_{jk} + A_{ki}$ is quantized, $F_{ijk} \in \{0, \pm N^2\}$, and the codifferential satisfies $d^*F = N\,A$, yielding a discrete Proca equation with mass parameter $m^2 = N$ (cf. the discrete gauge theory of[14]). Full proofs appear in the companion paper[16].

A. Numerical verification across primes

TABLE III. Verification of main identities across odd primes. All Frobenius norms $\|\cdot\|_F$ are below 10^{-10}, confirming exact identities to machine precision. The product $\prod s_k = N^{(N-2)/2}$ is verified to 14 significant digits.

N	$\|SA+AS\|_F$	$\|N^2S^2+A^2\|_F$	$\|L_+^2\|_F$	rank(L_+)
3	$< 10^{-15}$	$< 10^{-15}$	$< 10^{-15}$	1
5	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$	2
7	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	3
11	$< 10^{-12}$	$< 10^{-12}$	$< 10^{-12}$	5
13	$< 10^{-12}$	$< 10^{-11}$	$< 10^{-11}$	6
17	$< 10^{-11}$	$< 10^{-10}$	$< 10^{-10}$	8
37	$< 10^{-9}$	$< 10^{-8}$	$< 10^{-8}$	18

VI. NILPOTENT LIGHT-CONE STRUCTURE

A. Nilpotency and rank

Definition 26 (Light-cone operators)**.** For the deviated Qameah with parts S and A, define

$$L_\pm = S \pm \frac{1}{N} A. \tag{19}$$

Theorem 27 (Nilpotency)**.** $L_\pm^2 = 0$.

Proof. Expanding directly:

$$\begin{aligned} L_\pm^2 &= \left(S \pm \frac{1}{N}A\right)^2 = S^2 \pm \frac{1}{N}(SA + AS) + \frac{1}{N^2}A^2 \\ &= S^2 \pm \frac{1}{N}\{S, A\} + \frac{1}{N^2}A^2. \end{aligned} \tag{20}$$

By Theorem 16, $\{S, A\} = 0$, so the middle term vanishes. By Theorem 22, $A^2 = -N^2 S^2$, so

$$L_\pm^2 = S^2 + \frac{1}{N^2}(-N^2 S^2) = S^2 - S^2 = 0.$$

□

Theorem 28 (Maximal rank)**.** $\operatorname{rank}(L_+) = (N-1)/2$. *This is the maximum possible rank for a nilpotent-2 operator on $\mathbb{R}^N$ when N is odd.*

Proof. For a nilpotent-2 operator L on $\mathbb{R}^N$, $\operatorname{Im}(L) \subseteq \ker(L)$, so $\operatorname{rank}(L) \leq \dim\ker(L) = N - \operatorname{rank}(L)$, giving $\operatorname{rank}(L) \leq \lfloor N/2 \rfloor = (N-1)/2$ for odd N.

To show that L_+ achieves this bound, we work in the DFT basis. The operator L_+ decomposes into $(N-1)/2$ paired 2×2 blocks B_k (modes k and $N-k$) plus one zero mode ($k = 0$). At mode k:

$$B_k = \begin{pmatrix} s_k & a_k/N \\ -a_k/N & -s_k \end{pmatrix}$$

where $s_k = N/(2\sin(2\pi k/N))$ and $a_k = iNs_k$ (Proposition 21). Then $|a_k/N| = |s_k|$, so $\det(B_k) = -s_k^2 + s_k^2 = 0$ and $\operatorname{rank}(B_k) = 1$. Summing: $\operatorname{rank}(L_+) = (N-1)/2$.

Verified numerically for all odd primes $N = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$. □

Corollary 29 (Lagrangian subspace)**.** $\operatorname{Im}(L_+) \subseteq \ker(L_+)$ *with* $\dim\operatorname{Im}(L_+) = (N-1)/2$ *and* $\dim\ker(L_+) = (N+1)/2$. *The image is a maximal totally isotropic subspace of $\mathbb{R}^N$ under the nilpotent bilinear form $B(x, y) = x^T L_+ y$: a natural* Lagrangian subspace.

B. BRST cohomology

The nilpotent operator L_+ defines a cochain complex $0 \to \mathbb{R}^N \xrightarrow{L_+} \mathbb{R}^N \to 0$, and the cohomology $H^1(L_+) = \ker(L_+)/\operatorname{Im}(L_+)$ measures the space of "gauge-invariant" observables[4,9].

Theorem 30 (One-dimensional cohomology)**.** *For all standard Qameah of odd order N,*

$$\dim H^1(L_+) = \dim \frac{\ker(L_+)}{\operatorname{Im}(L_+)} = 1. \tag{21}$$

The unique cohomology class is generated by the all-ones vector $\mathbf{1} = (1, 1, \ldots, 1)^T$.

Proof. **Dimension count.** From Theorem 28, $\dim\ker(L_+) = (N{+}1)/2$ and $\dim\mathrm{Im}(L_+) = (N{-}1)/2$. Therefore $\dim H^1(L_+) = (N{+}1)/2 - (N{-}1)/2 = 1$.

Generator identification. We show $\mathbf{1} \in \ker(L_+) \setminus \mathrm{Im}(L_+)$.

- $\mathbf{1} \in \ker(L_+)$: the rows of $L_+ = S + A/N$ sum to zero. Indeed, S has zero row sums (deviated magic square) and A has zero row sums (antisymmetric matrix), so $L_+\,\mathbf{1} = \mathbf{0}$.
- $\mathbf{1} \notin \mathrm{Im}(L_+)$: the columns of L_+ also sum to zero (S has zero column sums by the magic property, and A has zero column sums since $A^T = -A$ with zero row sums). Therefore $\mathbf{1}^T L_+ = \mathbf{0}^T$, which means $\mathbf{1}$ is orthogonal to every vector in $\mathrm{Im}(L_+)$. Since $\mathbf{1} \neq \mathbf{0}$, it cannot lie in $\mathrm{Im}(L_+)$.

Hence $[\mathbf{1}]$ generates $H^1(L_+)$. Verified for $N = 3, 5, 7, \ldots, 37$ (13 values). □

Remark 31 (Quantum error-correcting codes). The nilpotency $L_+^2 = 0$ (Theorem 27) implies that the row space of $L_+ \bmod N$ is self-orthogonal over $\mathbb{F}_N$. By the CSS construction, this yields a quantum error-correcting code $[[N, N-4, 3]]_N$ for every prime $N \geq 5$. This code saturates the quantum Singleton bound and is therefore a quantum MDS code. The generator matrix is a Reed–Solomon $[N, 2, N-1]_N$ code, with the $\mathrm{GL}(2, \mathbb{Z}_N)$ magic square entries as evaluation points.

VII. THREE-GENERATION EQUIDISTRIBUTION FOR $N = 11$

A. Simplicial face counting

The $N{=}11$ Qameah acts on $\mathbb{R}^{11}$, whose standard basis vectors $\{e_0, e_1, \ldots, e_{10}\}$ span a 10-simplex Δ^{10}. The triangular (2-dimensional) faces of Δ^{10} are indexed by 3-element subsets of $\{0, 1, \ldots, 10\}$:

$$\binom{11}{3} = 165. \tag{22}$$

For each face $\{i, j, k\}$, define the *S-weight* as the sum of the absolute values of the relevant entries of S:

$$w_S(\{i, j, k\}) = 2(|S_{ij}| + |S_{ik}| + |S_{jk}|). \tag{23}$$

The total S-weight over all 165 faces is $W_{\mathrm{tot}} = 3 \times 20130 = 60390$.

B. Balanced partitions

Definition 32 (Balanced 3-partition)**.** A *balanced 3-partition* of the 165 faces is a decomposition $\mathcal{F} = \mathcal{G}_1 \sqcup \mathcal{G}_2 \sqcup \mathcal{G}_3$ with $|\mathcal{G}_g| = 55$ for each $g \in \{1, 2, 3\}$ and

$$\sum_{F \in \mathcal{G}_g} w_S(F) = 20130 \quad \text{for each } g. \tag{24}$$

Theorem 33 (Equidistribution)**.** *There exist exactly* $16 = 2^4$ *balanced 3-partitions of the 165 triangular faces of* Δ^{10} *satisfying* (24)*.*

The proof is constructive (exhaustive enumeration). The 165 faces decompose into orbits under the action of the cyclic group $\mathbb{Z}_{11}$ (generated by the index shift $i \mapsto i + 1 \bmod 11$). Each orbit has size 11, giving $165/11 = 15$ orbits. Of these:

- **5 self-conjugate orbits:** These are fixed under the complement map $\{i, j, k\} \mapsto \{N-i, N-j, N-k\}$. They form Generation 1 (55 faces).
- **10 paired orbits:** The remaining orbits come in 5 conjugate pairs under the complement map. Assigning one partner to Generation 2 and the other to Generation 3 gives a balanced partition, provided the S-weights match.
- 2^4 **choices:** Of the 5 conjugate pairs, 4 admit a free binary choice (either partner can go to either generation without breaking the weight balance), and 1 pair is fixed by the constraint. This gives $2^4 = 16$ solutions.

The enumeration code is available at `https://github.com/Insightful-Calculations/qhots_v7_calc`.

Proposition 34 (Edge balance)**.** *In every balanced 3-partition, each generation $\mathcal{G}_g$ contains exactly 55 edges (1-dimensional faces), and every edge participates in exactly $c_g(e) = 3$ triangular faces within its generation.*

Remark 35 (Random hit rate). A random 3-partition of 165 objects into groups of 55 achieves exact weight balance 20130/20130/20130 with probability $\approx 0.035\%$. The existence of 16 such partitions with additional edge-balance and orbit structure is a nontrivial combinatorial property of the N=11 Qameah.

Remark 36 (Universality for other N). Whether analogous balanced 3-partitions exist for other odd primes is an open question. The structure depends on the specific values of $\binom{N}{3}$, the orbit decomposition under $\mathbb{Z}_N$, and the S-weight distribution. The case N=11 is distinguished by $\binom{11}{3} = 165 = 3 \times 55$, where $55 = T_{10}$ (the 10th triangular number), enabling the threefold split. For N=7: $\binom{7}{3} = 35$, which is not divisible by 3, so no balanced 3-partition exists. For N=13: $\binom{13}{3} = 286$, which is not divisible by 3 either. Hence the three-generation structure is specific to N=11 among the first several primes.

VIII. SCOPE OF THE RESULTS

A. What is proved

1. **Anticommutation** $\{S, A\} = 0$ (Theorem 16): Proved algebraically for all standard Qameah of odd order, with the proof depending only on the complementarity of the circulant generator.
2. **Eigenvalue closed form** (Theorem 18): Proved via the Hankel–circulant factorization $S = J_{\text{rev}}\, C_h$ and standard DFT eigenvalue formulas. Universal for all odd N.
3. **Nilpotent identity** $N^2S^2 + A^2 = 0$ (Theorem 22): Proved mode-by-mode in the DFT basis. Universal for all odd N.
4. $L_\perp^2 = 0$ **and maximal rank** (Theorems 27–28): Direct consequences of (1)–(3). The rank bound is tight.
5. **BRST cohomology** $\dim H^1 = 1$ (Theorem 30): Follows from the dimension count. Universal for all odd N.
6. **Spectral generating function** (Theorem 8): Proved for all odd N; closes the spectral arithmetic series T91–T97.
7. **Three-generation equidistribution** (Theorem 32): Constructive proof by exhaustive enumeration, specific to N=11.

B. What is not proved

1. **No physical content is claimed.** The "Schrödinger dynamics" of Theorem 24 is a mathematical statement about the spectral decomposition of e^{-iSt}, not a claim about quantum mechanics. The "light-cone operators" $L_\pm$ are named by analogy; no physical light cone is implied.

2. **No coupling to physics.** Whether the algebraic structures identified here (anticommutation, nilpotency, BRST cohomology) have physical realizations in quantum field theory, condensed matter, or any other domain is an open question not addressed in this paper.

3. **No E8 connection is derived.** The coincidence that $(N^2-1)/4 = 30 = h(E_8)$ (the Coxeter number of the exceptional Lie algebra E_8) at N=11 is observed but not explained. It may be accidental.

4. **Strengthened divisibility.** The bound $N^{4k} \mid e_k$ (stronger than Theorem 3's $N^{2k} \mid e_k$) is verified only for N=11 and lacks a general proof.

5. **Composite N.** All theorems are stated for odd prime N. Composite odd N introduces additional divisibility structure that may require modifications to the generating function.

C. Negative results

1. Attempts to derive physical coupling constants from the eigenvalue ratios s_j/s_k found no matches beyond the level of numerical coincidence.

2. The Weinberg angle $\sin^2\theta_W$ cannot be obtained from any ratio of S-eigenvalues at N=11.

3. A discrete Proca Lagrangian $\mathcal{L} = N^5(N^2-1)/96$ with $d^*F = N\,A$ has been identified (Remark 25); the $\{S, A\} = 0$ relation itself is not yet derived from this action.

4. The three-generation structure does not extend to N=7 or N=13 (the adjacent primes), due to the divisibility constraint $3 \mid \binom{N}{3}$.

5. Although the anticommutation $\{S, A\} = 0$ is proved algebraically for all odd N (Theorem 16), analogous closed-form proofs of the strengthened spectral divisibility $N^{4k} \mid e_k$ have not been found.

IX. OPEN PROBLEMS

1. **Strengthened divisibility for general N.** Theorem 3 establishes $N^{2k} \mid e_k$ for all odd primes. The stronger bound $N^{4k} \mid e_k$, verified for $N = 11$, lacks a general proof. A p-adic or motivic approach may be required.

2. **Classification of nilpotent magic squares.** Non-Qameah members of the de la Loubère family violate $L_+^2 = 0$ for $N \geq 5$ (Section VI). A complete classification of which magic-square constructions yield the nilpotent light-cone structure is open. The relevant condition appears to be the complementarity of the circulant generator (Lemma 15).

3. **Composite odd N.** All theorems are stated for odd prime N. For composite N (e.g. $N = 9, 15, 21$), the DFT block structure changes and the generating function (Theorem 8) may require modification. Preliminary numerics suggest that anticommutation and nilpotency persist, but the rank formula changes.

4. **Three-generation structure beyond $N = 11$.** The balanced 3-partition of Theorem 32 is specific to $N = 11$. The obstruction for other primes is the divisibility $3 \mid \binom{N}{3}$, which fails for $N = 7, 13, 17$. Whether weaker forms of equidistribution exist is unexplored.

5. **From discrete Proca to $\{S, A\} = 0$.** A discrete Proca Lagrangian $\mathcal{L} = N^5(N^2-1)/96$ now exists (Remark 25), but the anticommutation $\{S, A\} = 0$ is not yet derived as its equation of motion. Whether the discrete Proca action *implies* $\{S, A\} = 0$ or requires additional structure remains open.

6. **Connections to representation theory.** The coincidence that each row of $|S|$ sums to $(N^2-1)/4$, which equals the Coxeter number $h(E_8) = 30$ at N=11, suggests a possible link to exceptional Lie algebras. No such connection has been established.

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